

INSIGHTBOOKS V2

Full Accounting Backend Reimplementation Plan

Product: InsightBooks Business Management System

Primary Objective: Rebuild the entire accounting backend so that every financial transaction is posted once, posted to the correct accounts, fully traceable, and consistently reflected in the General Ledger, Trial Balance, financial statements, operational reports, bank reconciliation, equity reports, accounting periods, and financial projections.

1. IMPLEMENTATION VISION

InsightBooks must be transformed from a collection of financial modules into a unified financial operating system.

The system currently contains accounting inconsistencies including:

- Unbalanced Trial Balance results.
- General Ledger balances that do not agree with reports.
- Duplicate financial postings.
- Transactions posted to incorrect accounts.
- Parent and child accounts being counted together incorrectly.
- Owner capital appearing more than once.
- Liabilities appearing in the Chart of Accounts but not in journals.
- Reports reading directly from operational tables.
- Different modules calculating their own independent balances.
- Missing or incomplete transaction audit trails.
- Posted transactions being editable without proper reversals.
- Financial reports that cannot be traced to their source transactions.

The reimplementation must establish one authoritative accounting path:

```
graph TD;
    A[Operational Transaction] --> B[Accounting Posting Engine];
    B --> C[Balanced Journal Entry];
    C --> D[Journal Entry Lines];
    D --> E[General Ledger];
    E --> F[Trial Balance];
    F --> G[Financial Statements];
```

Every financial module must use this same path.

The General Ledger must become the single source of truth for financial reporting. Operational modules such as sales, purchases, payroll, inventory, expenses, banking, loans, fixed assets, and equity must never update financial reports or account balances directly.

2. PROJECT OBJECTIVES

The implementation must accomplish the following objectives:

2.1 Accounting Integrity

- Every posted journal entry must balance.
- Total debits must always equal total credits.
- Every business transaction must create no more than one accounting event.
- Duplicate posting must be prevented at database and application levels.
- Every posting must use the correct Chart of Accounts mappings.
- Parent and child accounts must not be double-counted.
- Posted entries must become immutable.
- Corrections must be performed through reversals or adjustment journals.

2.2 Reporting Integrity

- The Trial Balance must be generated exclusively from posted journal lines.
- The General Ledger must be generated exclusively from posted journal lines.
- The Balance Sheet must be generated exclusively from the General Ledger.
- The Income Statement must be generated exclusively from the General Ledger.
- Cash Flow, equity, tax, aging, inventory, payroll, and management reports must reconcile with the General Ledger.
- Every report figure must show the account code and account name from which it was derived.
- Every figure must support drill-down to journal lines and source transactions.

2.3 Data Preservation

- Existing production data must not be deleted.
- Database migrations must be additive and reversible.
- Historical transactions must be analyzed before modification.
- Data repair scripts must be idempotent and safe to rerun.
- Every repair must be logged.
- A full backup and rollback strategy must be created before any production migration.

2.4 Operational Integration

The following modules must post through the centralized accounting engine:

- Sales
- Invoices
- Customer payments
- Credit notes
- Purchases
- Supplier bills
- Supplier payments
- Expenses
- Payroll
- Inventory
- Point of Sale
- Bank accounts
- Mobile money
- Cashbook
- Bank reconciliation
- Fixed assets
- Depreciation
- Asset disposal
- Loans
- Loan repayments
- Taxes
- Equity
- Owner drawings
- Dividends
- Opening balances
- Opening inventory
- Manual journals
- Year-end closing
- Period adjustments

2.5 Auditability

Every financial transaction must be traceable from:

Financial Statement
→ Report Line
→ Ledger Account
→ Journal Entry Line
→ Journal Entry
→ Source Transaction
→ Source Document
→ User and Approval History

3. IMPLEMENTATION PRINCIPLES

The development team must follow these principles throughout the project.

3.1 General Ledger as the Single Source of Truth

Financial statements must never calculate balances directly from:

- Invoice tables
- Payment tables
- Sales tables
- Expense tables
- Inventory tables
- Payroll tables
- Asset tables
- Bank transaction tables
- Loan tables
- Equity transaction tables

These operational tables may provide supporting information, but financial values must come from posted journal entries.

3.2 One Accounting Event, One Posting

Each financial event must have a unique accounting identity.

Examples:

```
Invoice ID + POST_INVOICE  
Payment ID + POST_CUSTOMER_PAYMENT  
Expense ID + POST_EXPENSE  
Payroll Run ID + POST_PAYROLL  
Asset ID + POST_ASSET_ACQUISITION
```

The system must reject attempts to post the same event more than once.

3.3 Balanced Entries

Before posting:

```
Sum of Debits = Sum of Credits
```

Any unbalanced entry must be rejected before the database transaction is committed.

3.4 Atomic Posting

The following actions must occur inside one database transaction:

1. Validate the source transaction.
2. Resolve account mappings.
3. Create the journal entry.
4. Create journal lines.
5. Mark the source transaction as posted.
6. Create audit records.
7. Update any rebuildable reporting projections or caches.
8. Commit everything together.

If any step fails, the entire transaction must roll back.

3.5 Immutability

Once posted:

- Journal dates cannot be silently changed.
- Accounts cannot be changed.
- Debit and credit values cannot be edited.
- Journal lines cannot be deleted.
- Source linkage cannot be changed.
- Posted source transactions cannot be financially edited.

Corrections must use:

- Full reversal
- Partial reversal where legally and structurally appropriate
- Adjustment journal
- Credit note
- Debit note
- Void-and-repost workflow

3.6 Exact Monetary Values

All monetary values must use exact decimal database types.

Do not use:

- JavaScript floating-point calculations for accounting totals.
- Database `FLOAT` or `REAL` types for financial amounts.
- Rounded intermediate values without defined precision rules.

Use controlled decimal precision and currency rounding policies.

3.7 Separation of Actuals and Forecasts

Actual accounting data and financial projections must remain separate.

Actual Data → General Ledger

Forecast Data → Projection Models

Forecasts must never post to the actual General Ledger.

4. PROJECT DELIVERY STRATEGY

The system must not be rewritten in one uncontrolled release.

Implementation must be divided into controlled phases.

Each phase must include:

- Repository analysis.
- Database analysis.
- Technical design.
- Implementation.
- Automated testing.
- Data validation.
- User acceptance testing.
- Migration scripts.
- Rollback scripts.
- Technical documentation.
- Deployment checklist.
- Sign-off criteria.

The recommended implementation order is:

Phase 0 – Project Safety and Environment Preparation
Phase 1 – Complete Accounting Forensic Audit
Phase 2 – Target Accounting Architecture
Phase 3 – Chart of Accounts Reconstruction
Phase 4 – Accounting Posting Engine
Phase 5 – Journal and General Ledger Reimplementation
Phase 6 – Historical Data Reconciliation and Repair
Phase 7 – Trial Balance and Financial Reports
Phase 8 – Financial Calendar and Accounting Periods
Phase 9 – Operational Module Integration
Phase 10 – Bank Reconciliation
Phase 11 – Equity Management
Phase 12 – Period Closing and Year-End Closing
Phase 13 – Financial Planning and Projection
Phase 14 – Loan Readiness and Executive Reporting
Phase 15 – Security, Audit, and Compliance
Phase 16 – Performance and Scalability

5. PHASE 0 — PROJECT SAFETY AND ENVIRONMENT PREPARATION

5.1 Establish Environments

Maintain separate environments for:

- Local development
- Shared development
- Testing
- Staging
- User acceptance testing
- Production

No accounting repair script should be executed directly in production before being validated against a recent production database copy.

5.2 Create Full Backups

Before any database changes:

- Create a full PostgreSQL custom-format backup.
- Back up uploaded files and transaction attachments.
- Back up environment configuration securely.
- Record application version and Git commit.
- Record database migration version.
- Validate that the backup can be restored successfully.

A backup is not considered valid until a restoration test has passed.

5.3 Establish a Migration Ledger

Create an internal migration registry containing:

- Migration name
- Migration version
- Date executed
- Environment
- User or deployment process
- Number of records inspected
- Number of records changed
- Number of records skipped
- Errors

- Rollback status
- Integrity-check results

5.4 Freeze Unsafe Accounting Changes

During the initial audit, prevent developers from introducing new direct balance updates.

Create a temporary development rule:

No module may add or modify accounting behaviour until the existing posting flow has been documented.

5.5 Establish Baseline Metrics

Capture the current state:

- Total journals.
- Total journal lines.
- Total unbalanced journals.
- Total source transactions without journals.
- Total journals without sources.
- Total duplicate postings.
- Total accounts with invalid categories.
- Total transactions posted to inactive accounts.
- Total periods represented in the data.
- Current Trial Balance difference.
- Balance Sheet imbalance.
- General Ledger and report differences.
- Number of negative or abnormal account balances.
- Number of owner capital discrepancies.
- Number of liabilities without traceable journal lines.

6. PHASE 1 — COMPLETE ACCOUNTING FORENSIC AUDIT

Cursor must analyze the complete application before implementing fixes.

6.1 Repository Audit

Inspect:

- Database schema.
- Prisma schema or ORM models.
- Migrations.
- API routes.
- Server actions.
- Controllers.

- Services.
- Background jobs.
- Event listeners.
- Webhooks.
- Scheduled jobs.
- Import scripts.
- Seeders.
- Report queries.
- Dashboard queries.
- Utility functions.
- Account balance calculations.
- Journal creation functions.
- Account mapping constants.
- Source transaction status changes.
- Reversal logic.
- Delete and update handlers.

6.2 Identify Every Posting Path

Produce a posting inventory such as:

Module	Transaction	Existing Posting Function	Current Debit	Current Credit	Duplicate Risk	Status
Sales	Invoice	Existing service	Receivable	Revenue/VAT	High	Review
Sales	Customer payment	API handler	Bank	Receivable	Medium	Review
Expenses	Expense	Multiple handlers	Expense	Cash/ Payable	High	Review
Payroll	Payroll run	Payroll service	Salaries	PAYE/ Pension/ Bank	High	Review

No financial transaction type should be omitted.

6.3 Detect Duplicate Posting Causes

Search for:

- API route posting plus service-layer posting.
- `afterCreate` or event listeners repeating API posting.
- Webhooks repeating completed payments.
- Retry processes without idempotency.
- Frontend double submission.
- Multiple posting functions for the same transaction.
- Posting during record creation and again during approval.
- Imports posting data that has already been posted.

- Reconciliation adjustments creating repeated journals.
- Reports summing both source transactions and journals.
- Parent accounts and child accounts aggregated together.

6.4 Detect Wrong Account Mappings

Review:

- Hardcoded account IDs.
- Hardcoded account names.
- Account-code fallbacks.
- Default account creation.
- Per-business mappings.
- Missing account mappings.
- Duplicate account codes.
- Accounts sharing the same purpose.
- Transactions posted to parent accounts instead of posting accounts.
- Expense categories mapped outside the expense hierarchy.
- Salaries posted to multiple salary accounts.
- Taxes posted directly to expenses instead of liability accounts.
- Loan principal posted as income or expense.
- Asset purchases posted entirely as expenses.
- Capital contributions posted as revenue.
- Owner withdrawals posted as business expenses.

6.5 Analyze Historical Data

Generate reports for:

- Duplicate journal entries.
- Identical journal lines.
- Multiple journals linked to one source.
- Sources linked to multiple active journals.
- Journals with no source.
- Sources marked posted without journals.
- Journals marked posted without lines.
- Journals with unequal debits and credits.
- Lines with both debit and credit values.
- Lines with neither debit nor credit.
- Missing account references.
- Deleted or inactive account references.
- Cross-business posting.
- Cross-branch posting.
- Transactions outside valid financial years.
- Backdated transactions in closed periods.
- Reversals without original journals.
- Original journals reversed more than once.

6.6 Audit Deliverable

Produce a formal audit report containing:

- Current architecture.
- Identified posting paths.
- Accounting defects.
- Duplicate-posting causes.
- Wrong-account causes.
- Report-source inconsistencies.
- Data corruption risks.
- Historical-data repair categories.
- Recommended migration strategy.
- High-risk modules.
- Dependencies.
- Proposed implementation order.

No destructive repair should begin until this audit is completed.

7. PHASE 2 — TARGET ACCOUNTING ARCHITECTURE

7.1 Core Components

Implement or reimplement:

- Chart of Accounts
- Account mappings
- Accounting Posting Engine
- Posting templates
- Journal entries
- Journal lines
- Financial years
- Accounting periods
- General Ledger query service
- Trial Balance service
- Reversal service
- Adjustment service
- Period closing service
- Year-end closing service
- Audit trail
- Attachments
- Approval workflow
- Report engine
- Accounting integrity service

7.2 Core Service Boundaries

Recommended services:

```
AccountingPostingService
AccountMappingService
JournalValidationService
JournalPostingService
JournalReversalService
LedgerQueryService
TrialBalanceService
FinancialStatementService
PeriodService
PeriodClosingService
YearEndClosingService
BankReconciliationService
EquityPostingService
AccountingAuditService
HistoricalRepairService
```

7.3 Posting Command Structure

Every posting request should follow a standard structure:

```
Posting Command
- businessId
- sourceModule
- sourceType
- sourceId
- eventType
- transactionDate
- postingDate
- currency
- exchangeRate
- branchId
- departmentId
- projectId
- costCentreId
- description
- userId
- approvalId
- attachmentIds
- metadata
```

7.4 Posting Result

Every successful posting should return:

- journalEntryId
- journalNumber
- postingDate
- accountingPeriodId
- totalDebit
- totalCredit
- status
- sourceReference
- auditReference

8. PHASE 3 — CHART OF ACCOUNTS RECONSTRUCTION

8.1 Account Structure

Every account must include:

- Business or tenant
- Account code
- Account name
- Description
- Parent account
- Account category
- Account type
- Normal balance
- Financial statement section
- Cash-flow classification
- Tax mapping
- Currency policy
- Posting permission
- Active status
- System account flag
- Created and updated audit fields

8.2 Account Categories

At minimum:

- Assets
- Liabilities
- Equity
- Revenue
- Cost of Sales
- Expenses
- Other Income
- Other Expenses

8.3 Posting and Control Accounts

Differentiate between:

- Header accounts
- Parent accounts
- Posting accounts
- Control accounts
- System accounts
- Contra accounts

Transactions must not be posted to non-posting parent accounts.

8.4 Account Uniqueness

Enforce:

- Unique account codes per business.
- No duplicate active system-account mappings.
- One account per required accounting purpose unless explicitly designed otherwise.
- No repeated salary, revenue, receivable, payable, tax, capital, or retained-earnings mappings.

8.5 Expense Account Rules

The expense-entry interface must retrieve categories only from active posting accounts under the configured Expenses hierarchy.

Salaries and wages must post only to the approved Salaries and Wages account, such as Account 5200, unless the business has deliberately configured a valid replacement.

Remove or merge conflicting accounts such as:

- Salaries Expense
- Salary Costs
- Payroll Expense
- Wages Expense

Historical records must be remapped carefully rather than deleted.

8.6 Parent and Child Aggregation

A financial report must use one of two methods:

1. Sum posting accounts and display parent accounts as presentation totals.
2. Sum only the direct balances of each selected account.

It must never sum parent totals and then add child totals again.

8.7 Account Mapping Registry

Create a business-specific mapping table for:

- Accounts Receivable
- Accounts Payable
- Sales Revenue
- Service Revenue
- Inventory
- Cost of Sales
- VAT Output
- VAT Input
- Cash
- Bank
- Mobile Money
- Salaries and Wages
- PAYE Payable
- Pension Payable
- Loan Principal
- Interest Expense
- Fixed Assets
- Accumulated Depreciation
- Depreciation Expense
- Owner Capital
- Share Capital
- Retained Earnings
- Current Year Profit
- Owner Drawings
- Dividends Payable
- Bank Charges
- Exchange Gains and Losses
- Opening Balance Equity

9. PHASE 4 — CENTRAL ACCOUNTING POSTING ENGINE

9.1 Responsibilities

The engine must:

1. Receive a posting command.
2. Validate business ownership.
3. Validate transaction status.
4. Validate approval status.
5. Resolve accounting period.
6. Confirm the period is open.
7. Resolve account mappings.

8. Generate journal lines.
9. Validate debit-credit equality.
10. Validate currencies and exchange rates.
11. Create the journal and journal lines.
12. Link the journal to its source.
13. Mark the source as posted.
14. Write audit records.
15. Return the posting result.

9.2 Idempotency

Create a unique constraint similar to:

```
businessId
+ sourceModule
+ sourceType
+ sourceId
+ eventType
+ activePostingStatus
```

Retries must return the original journal result rather than generating another journal.

9.3 Journal Numbering

Generate controlled journal numbers such as:

```
JE-2026-000001
SALE-2026-000001
PAY-2026-000001
EXP-2026-000001
PAYROLL-2026-000001
BANK-2026-000001
EQ-2026-000001
```

Number generation must be safe under concurrent requests.

9.4 Validation Rules

Reject postings where:

- Debits do not equal credits.
- Accounts belong to another business.
- An account is inactive.
- A parent account is non-posting.
- A period is closed.
- Required mappings are missing.
- Monetary values are invalid.
- Source records do not exist.

- The source is cancelled or deleted.
- The event has already been posted.
- Required approval has not been completed.
- Currency information is inconsistent.
- Dimensions conflict with source data.

9.5 Posting Templates

Use configurable posting templates.

Example:

Customer Invoice

Dr Accounts Receivable
Cr Sales Revenue
Cr VAT Output, where applicable

Customer Payment

Dr Bank/Cash/Mobile Money
Cr Accounts Receivable

Cash Expense

Dr Expense Account
Dr VAT Input, where applicable
Cr Bank/Cash/Mobile Money

Supplier Bill

Dr Expense/Inventory/Asset
Dr VAT Input, where applicable
Cr Accounts Payable

Supplier Payment

Dr Accounts Payable
Cr Bank/Cash

10. PHASE 5 — JOURNAL AND GENERAL LEDGER REIMPLEMENTATION

10.1 Journal Entry Model

Include:

- Journal ID
- Business ID
- Journal number
- Source module
- Source type
- Source ID
- Event type
- Transaction date
- Posting date
- Financial year
- Accounting period
- Description
- Currency
- Exchange rate
- Status
- Approval status
- Reversal status
- Original journal ID
- Created by
- Approved by
- Posted by
- Created timestamp
- Posted timestamp
- Metadata

10.2 Journal Line Model

Include:

- Journal line ID
- Journal entry ID
- Account ID
- Debit
- Credit
- Base-currency debit
- Base-currency credit
- Description
- Customer
- Supplier
- Employee
- Owner
- Bank account

- Branch
- Department
- Project
- Cost centre
- Tax code
- Due date
- Reference
- Sequence

10.3 General Ledger Architecture

The authoritative ledger must be derived from posted journal lines.

Do not maintain an independently editable General Ledger table.

A materialized view or cache may be used for performance only when:

- It can be fully rebuilt from journal lines.
- It is never treated as the source of truth.
- Rebuild procedures are available.
- Integrity checks compare it with journal lines.

10.4 Ledger Views

Support:

- Account ledger
- Business ledger
- Branch ledger
- Department ledger
- Project ledger
- Cost-centre ledger
- Customer subledger
- Supplier subledger
- Employee subledger
- Owner subledger
- Bank-account ledger

10.5 Opening and Closing Balances

For any selected range:

```
Opening Balance
+ Period Debits
- Period Credits
= Closing Balance
```

The presentation must respect each account's normal balance.

10.6 Journal Reversal

A reversal must:

- Create a new journal.
 - Reverse every original debit and credit.
 - Link to the original journal.
 - Record the reason.
 - Record the authorized user.
 - Prevent repeated active reversals.
 - Respect accounting-period permissions.
 - Preserve both journals permanently.
-

11. PHASE 6 — HISTORICAL DATA RECONCILIATION AND REPAIR

Historical repair must be evidence-driven.

11.1 Classify Historical Records

Place discrepancies into categories:

- Valid posting
- Duplicate journal
- Missing journal
- Unbalanced journal
- Wrong account
- Wrong amount
- Wrong date
- Wrong business
- Wrong period
- Missing source
- Orphaned source
- Invalid reversal
- Duplicate opening balance
- Duplicate capital posting
- Liability without journal support
- Parent-child double count
- Report-only discrepancy

11.2 Duplicate Detection

Use combinations of:

- Source record
- Event type
- Amount

- Date
- Reference
- Account pairs
- User
- Posting time
- External payment reference
- Import batch

Never delete duplicates automatically based only on matching amounts.

11.3 Historical Repair Actions

Depending on evidence:

- Link an orphan journal to the correct source.
- Create a missing journal.
- Reverse a duplicate journal.
- Create an adjustment journal.
- Remap an account.
- Correct report mappings.
- Consolidate duplicate accounts.
- Reclassify account types.
- Rebuild reporting caches.
- Reassign accounting periods.
- Mark irreparable records for manual review.

11.4 Owner Capital Repair

Investigate the reported case where MK1,000,000 was posted but MK2,000,000 appeared.

Check for:

- Duplicate journals.
- Dashboard summing source and journal data.
- Parent and child equity double counting.
- Current-year profit incorrectly added to capital.
- Opening balance plus capital contribution duplication.
- Owner subledger and equity account both being aggregated.
- Stored account balance plus journal-derived balance.

Repair must preserve the correct MK1,000,000 accounting event and reverse or exclude the duplicated impact.

11.5 Liability Repair

For liabilities appearing in the Chart of Accounts but not journals:

- Determine whether the Chart of Accounts view is showing stored balances.
- Identify source obligations.
- Locate missing supplier, tax, loan, or payroll journals.

- Create controlled migration journals where supporting evidence exists.
- Do not invent liabilities merely to force reports to balance.
- Flag unsupported balances for finance-team review.

11.6 Repair Logging

Every repair record must capture:

- Original values
- Corrected values
- Reason
- Evidence
- Script version
- User or migration
- Timestamp
- Related journal
- Rollback procedure

12. PHASE 7 — TRIAL BALANCE AND FINANCIAL REPORTS

12.1 Trial Balance

The Trial Balance must:

- Read posted journal lines only.
- Support date and accounting-period filters.
- Show opening balance.
- Show period debits.
- Show period credits.
- Show closing debit or credit balance.
- Show account code and name.
- Support account hierarchy.
- Avoid parent-child double counting.
- Support branch, department, project, and cost-centre filters.
- Confirm total debits equal total credits.

The system must block final financial-statement approval if the Trial Balance does not balance.

12.2 General Ledger Report

Each account must show:

- Opening balance
- Transaction date
- Posting date
- Journal number

- Source module
- Source reference
- Description
- Debit
- Credit
- Running balance
- User
- Approval status
- Attachments
- Reversal status

12.3 Income Statement

Structure:

```
Revenue
Less: Cost of Sales
Gross Profit
Less: Operating Expenses
EBITDA
Less: Depreciation and Amortization
Operating Profit
Less: Finance Costs
Profit Before Tax
Less: Tax
Net Profit
```

12.4 Balance Sheet

Structure:

```
Assets
  Current Assets
  Non-current Assets

Liabilities
  Current Liabilities
  Long-term Liabilities

Equity
  Share Capital
  Capital Contributions
  Retained Earnings
  Current Year Profit
  Drawings
```

The report must always validate:

Assets = Liabilities + Equity

12.5 Cash Flow Statement

Implement cash flow classifications:

- Operating activities
- Investing activities
- Financing activities

The report must reconcile opening cash, movement, and closing cash with cash and bank ledger accounts.

12.6 Statement of Changes in Equity

Show:

- Opening equity
- Capital contributions
- Share issues
- Current-year profit or loss
- Dividends
- Drawings
- Prior-period adjustments
- Closing equity

12.7 Report Drill-Down

Every line must support:

Report line
→ Accounts included
→ Journal lines
→ Journal entry
→ Source transaction
→ Source document

12.8 Report Exports

Support:

- PDF
- Excel
- Print
- Company branding
- Reporting period
- Generation date

- Prepared by
 - Reviewed by
 - Approval section
 - Account codes and names
 - Comparative periods
-

13. PHASE 8 — FINANCIAL CALENDAR AND ACCOUNTING PERIODS

Every transaction must automatically receive a financial year and accounting period based on its transaction date. Users should not ordinarily select accounting periods manually.

13.1 Financial Year Setup

Store:

- Name
- Start date
- End date
- Status
- Current active year
- Number of periods
- Business ID

Support financial years such as:

- January to December
- July to June
- Other valid twelve-month configurations

13.2 Automatic Period Creation

When a financial year is created:

- Generate monthly periods automatically.
- Assign correct start and end dates.
- Account for leap years.
- Prevent overlapping periods.
- Prevent gaps.
- Set initial status.

13.3 Period Statuses

Use:

- Open
- Closed

- Reopened

13.4 Period Closing Checklist

Before closure, verify:

- All journals balance.
- No required transactions remain unposted.
- Bank reconciliation is complete or exceptions are authorized.
- Inventory is updated.
- Payroll is completed.
- Taxes are recorded.
- Trial Balance balances.
- Required approvals are complete.
- Suspense accounts are reviewed.
- Draft journals are resolved.
- Reconciliation differences are resolved.
- Required reports have been generated.

13.5 Closed Period Controls

Normal users cannot:

- Create backdated transactions.
- Edit posted transactions.
- Delete transactions.
- Post adjustments.
- Change journal dates.

Authorized corrections must use controlled adjustments or a formal reopening workflow.

13.6 Reopening

Reopening requires:

- Authorized role.
 - Reason.
 - Approval.
 - Audit record.
 - Reopening date.
 - Automatic notification.
 - Re-closing checklist.
-

14. PHASE 9 — OPERATIONAL MODULE INTEGRATION

14.1 Sales and Invoicing

Invoice

Dr Accounts Receivable
Cr Sales or Service Revenue
Cr VAT Output

Cash Sale

Dr Cash/Bank/Mobile Money
Cr Sales Revenue
Cr VAT Output

Customer Payment

Dr Cash/Bank/Mobile Money
Cr Accounts Receivable

Customer Credit Note

Dr Sales Returns
Dr VAT Output Adjustment
Cr Accounts Receivable

Prevent invoice posting from being repeated when:

- An invoice is approved.
- An invoice is emailed.
- A payment is recorded.
- A receipt is generated.
- A webhook confirms payment.

14.2 Purchases and Payables

Supplier Bill

Dr Expense/Inventory/Asset
Dr VAT Input
Cr Accounts Payable

Supplier Payment

Dr Accounts Payable
Cr Bank/Cash

Supplier Credit

Dr Accounts Payable
Cr Expense/Inventory/Asset
Cr VAT Input Adjustment

14.3 Expenses

Support:

- Immediate payment
- Unpaid expense
- Reimbursement
- Petty cash
- Mobile money
- Recurring expense
- Staff expense claim
- Supplier-linked expense

The category must resolve to a valid posting account under Expenses.

14.4 Inventory

Inventory Purchase

Dr Inventory
Cr Accounts Payable/Bank

Inventory Sale

Revenue entry:

Dr Accounts Receivable/Bank
Cr Sales Revenue
Cr VAT Output

Cost entry:

Dr Cost of Sales
Cr Inventory

Inventory valuation and cost-of-sales policies must be consistent.

14.5 Payroll

Typical posting:

Dr Salaries and Wages
Dr Employer Pension Expense
Cr PAYE Payable
Cr Employee Pension Payable
Cr Employer Pension Payable
Cr Salary Deductions Payable
Cr Bank/Payroll Payable

Salary costs must not be posted to multiple expense accounts.

14.6 Fixed Assets

Acquisition

Dr Fixed Asset
Cr Bank/Accounts Payable

Depreciation

Dr Depreciation Expense
Cr Accumulated Depreciation

Disposal

Record:

- Asset cost removal
- Accumulated depreciation removal
- Proceeds
- Profit or loss on disposal

14.7 Loans

Loan Proceeds

```
Dr Bank
Cr Loan Liability
```

Loan Repayment

```
Dr Loan Liability
Dr Interest Expense
Cr Bank
```

Loan principal must not be treated as revenue or expense.

14.8 Taxes

Maintain separate accounts for:

- VAT Output
- VAT Input
- VAT Payable/Receivable
- PAYE Payable
- Withholding Tax
- Corporate Tax
- Other statutory obligations

14.9 Opening Balances

Opening balances must be posted through a controlled opening journal.

Assets, liabilities, and equity must be supported by evidence.

The opening journal must balance before posting.

15. PHASE 10 — BANK RECONCILIATION

The Bank Reconciliation module must compare the book balance recorded in InsightBooks with imported bank statement activity.

15.1 Bank Account Setup

Store:

- Bank name

- Branch
- Account name
- Account number
- Currency
- Linked General Ledger account
- Opening balance
- Opening date
- Status
- Business
- Branch or department where applicable

15.2 Statement Import

Support:

- CSV
- Excel
- Future OFX
- Future direct-bank APIs

Validate:

- Required columns
- Dates
- Debit and credit values
- Running balances
- Duplicate file imports
- Duplicate statement lines
- Currency
- Account ownership
- Statement period

15.3 Matching Engine

Match using:

- Exact amount
- Date tolerance
- Reference
- Description
- Payee or payer
- Transaction type
- External payment reference
- Batch settlement reference

Assign confidence scores:

- Exact
- High confidence
- Possible
- No match

15.4 Matching Types

Support:

- One bank line to one book transaction.
- One bank line to multiple book transactions.
- Multiple bank lines to one book transaction.
- Split transactions.
- Merged settlements.
- Partial matches where properly controlled.

15.5 Reconciliation Adjustments

Support controlled posting of:

- Bank charges
- Interest income
- Direct debits
- Direct deposits
- Standing orders
- Loan repayments
- Returned payments
- Mobile money settlements
- Currency adjustments

Each adjustment must create a balanced journal through the Accounting Posting Engine.

15.6 Completion

A reconciliation can be completed only when:

- The statement closing balance is recorded.
- The adjusted book balance agrees with the bank balance.
- All differences are explained.
- Outstanding items are documented.
- Required approval is complete.

15.7 Reversal

Reversing a reconciliation must:

- Preserve the original record.
 - Record a reason.
 - Reverse reconciliation-created adjustments where appropriate.
 - Reopen affected items.
 - Maintain the audit trail.
-

16. PHASE 11 — EQUITY MANAGEMENT

The Equity module must support capital contributions, share capital, retained earnings, owner drawings, dividends, and ownership registers while hiding journal-entry complexity from ordinary users.

16.1 Owners and Shareholders

Store:

- Full name
- Type
- Contact details
- Ownership percentage
- Date joined
- Status
- Number of shares
- Nominal value
- Capital account
- Drawings account
- Dividend information

16.2 Capital Contributions

Cash Contribution

```
Dr Bank/Cash
Cr Owner Capital/Share Capital
```

Asset Contribution

```
Dr Relevant Asset
Cr Owner Capital/Capital Contribution
```

Capital contributions must not automatically change ownership percentages.

16.3 Share Issues

```
Dr Bank/Shareholder Receivable
Cr Share Capital
Cr Share Premium
```

16.4 Owner Drawings

Dr Owner Drawings
Cr Bank/Cash/Relevant Asset

Owner drawings must reduce equity and must not appear as normal business expenses.

16.5 Dividend Declaration

Dr Retained Earnings/Dividends Declared
Cr Dividends Payable

16.6 Dividend Payment

Dr Dividends Payable
Cr Bank

16.7 Current-Year Profit

Current-year profit must be calculated from posted revenue and expense account balances.

It must not be manually entered as an independent balance.

16.8 Equity Reports

Implement:

- Equity dashboard
- Equity Statement
- Statement of Changes in Equity
- Owner statement
- Capital contributions report
- Share-capital report
- Dividend report
- Owner-drawings report

17. PHASE 12 — PERIOD CLOSING AND YEAR-END CLOSING

17.1 Month-End Closing

Generate a close checklist and store each result.

Require approval according to role.

17.2 Year-End Closing

The year-end process must:

1. Validate all periods.
2. Generate final Trial Balance.
3. Generate financial statements.
4. Calculate annual profit or loss.
5. Close revenue accounts.
6. Close expense accounts.
7. Transfer profit or loss to retained earnings.
8. Carry forward assets.
9. Carry forward liabilities.
10. Carry forward equity.
11. Create the next financial year.
12. Generate new accounting periods.
13. Store year-end closing journals.
14. Lock the completed year.

17.3 Closing Entries

Closing journals must be identifiable, auditable, and reversible only under highly restricted authorization.

18. PHASE 13 — FINANCIAL PLANNING AND PROJECTION

The Financial Planning module must use trusted closed-period historical data to prepare forecasts, scenarios, trends, and loan-readiness analysis.

This phase must begin only after historical accounting information is reliable.

18.1 Projection Data Sources

Use:

- Closed accounting periods.
- Historical General Ledger data.
- Sales trends.
- Expense behaviour.
- Customer collection trends.
- Supplier payment trends.
- Inventory turnover.
- Loan schedules.
- Payroll commitments.

- Asset plans.
- Tax assumptions.
- Seasonal trends.

18.2 Assumptions Engine

Support:

- Revenue growth
- Seasonal adjustment
- Customer growth
- Customer churn
- Product pricing
- Inflation
- Salary increases
- Rent increases
- Utility increases
- Marketing plans
- Capital expenditure
- Asset disposal
- New loans
- Interest rates
- Repayment schedules
- Tax rates
- Inventory cost inflation
- Stock turnover

Every assumptions change must trigger a recalculation of dependent projections.

18.3 Projection Types

Implement:

- Projected Profit and Loss
- Projected Cash Flow
- Projected Balance Sheet
- Working Capital forecast
- Loan capacity
- Cash runway
- EBITDA projection
- Debt-ratio projection
- Retained-earnings projection

18.4 Scenario Planning

Support:

- Conservative
- Expected
- Optimistic

- Custom scenarios

Scenarios must be isolated and versioned.

18.5 Projected Balance Sheet Integrity

Every projected Balance Sheet must validate:

Projected Assets
=
Projected Liabilities
+
Projected Equity

18.6 Variance Analysis

Compare:

Projected
vs
Actual

Show:

- Amount
- Variance
- Percentage
- Status
- Explanation
- Drill-down to actual transactions

18.7 AI-Assisted Forecasting

The AI feature may:

- Analyze historical trends.
- Prepare draft assumptions.
- Identify seasonality.
- Identify unusual expenses.
- Explain projected cash shortages.
- Produce plain-language summaries.

AI output must be:

- Clearly marked as a draft.
- Reviewable.
- Editable.
- Versioned.

- Separate from actual accounting records.
 - Unsupported by automatic General Ledger postings.
-

19. PHASE 14 — LOAN READINESS AND EXECUTIVE REPORTING

19.1 Ratios

Implement:

- Current ratio
- Quick ratio
- Gross profit margin
- Net profit margin
- Debt-to-equity ratio
- Interest-coverage ratio
- Debt-service coverage ratio
- Working capital
- Return on assets
- Inventory turnover
- Receivable collection period
- Payable payment period

19.2 Readiness Checks

Check:

- Financial statements generated.
- Trial Balance balanced.
- Accounting periods closed.
- Bank reconciliations completed.
- Tax reports available.
- Audit trail completeness.
- Cash flow stability.
- Revenue consistency.
- Profitability.
- Debt capacity.
- Supporting documents.

Include a clear disclaimer that the result is an internal financial-readiness assessment and not a guarantee of financing.

20. PHASE 15 — SECURITY, APPROVALS, AND AUDIT

20.1 Roles

Support at least:

- Super Administrator
- Administrator
- Business Owner
- CEO
- Finance Manager
- Accountant
- Cashier
- Payroll Officer
- Auditor
- Read-only finance user

20.2 Permissions

Control:

- Create draft transactions
- Approve transactions
- Post journals
- Reverse journals
- Create adjustments
- Close periods
- Reopen periods
- Run year-end closing
- Import bank statements
- Match bank transactions
- Complete reconciliations
- Reverse reconciliations
- Configure account mappings
- Manage Chart of Accounts
- Record equity transactions
- Approve equity transactions
- View financial reports
- Export reports
- View audit trails

20.3 Audit Trail

Capture:

- User
- Action

- Date and time
- Previous values
- New values
- Source record
- Journal entry
- Device information where available
- IP address where permitted
- Approval information
- Reason
- Attachments

20.4 Auditor Role

Auditors should be able to:

- View reports.
- View attachments.
- Trace transactions.
- Review journals.
- Review reversals.
- Review period closing.
- Review reconciliations.
- Export audit reports.
- View immutable audit history.

Auditors should not be able to alter accounting data unless separately granted an operational role.

21. PHASE 16 — AUTOMATED ACCOUNTING TESTING

21.1 Unit Tests

Test:

- Account mapping.
- Debit-credit generation.
- Decimal calculations.
- Journal validation.
- Period resolution.
- Currency conversion.
- Idempotency.
- Reversals.
- Account normal balances.
- Parent-child aggregation.
- Ratio calculations.

21.2 Integration Tests

For every module:

```
Source transaction
→ Posting engine
→ Journal
→ Ledger
→ Trial Balance
→ Financial statements
```

21.3 Double-Entry Tests

For every test transaction:

```
Total Debit = Total Credit
```

21.4 Report Reconciliation Tests

Validate:

- General Ledger equals Trial Balance.
- Trial Balance debit total equals credit total.
- Income Statement net profit agrees with equity movement.
- Balance Sheet balances.
- Cash Flow closing cash agrees with cash and bank accounts.
- Receivable aging agrees with Accounts Receivable control account.
- Payable aging agrees with Accounts Payable control account.
- Inventory valuation agrees with Inventory control account.
- Asset register agrees with fixed-asset accounts.
- Loan schedules agree with loan liabilities.
- Equity reports agree with equity accounts.

21.5 Duplicate Prevention Tests

Simulate:

- Double clicks.
- API retries.
- Network retries.
- Duplicate webhooks.
- Repeated imports.
- Concurrent posting.
- Multiple workers.
- Duplicate approval callbacks.

Only one active journal may be produced.

21.6 Closed Period Tests

Ensure:

- Normal posting is rejected.
- Unauthorized edits are rejected.
- Reversals follow permission rules.
- Reopening is audited.
- Re-closing performs validation again.

21.7 Migration Tests

Test repair scripts using:

- Clean database.
 - Production-like database.
 - Partially migrated database.
 - Script rerun.
 - Failed execution.
 - Rollback.
 - Large datasets.
-

22. PHASE 17 — PERFORMANCE AND SCALABILITY

22.1 Database Indexes

Index:

- Business ID
- Account ID
- Journal ID
- Source ID
- Posting date
- Transaction date
- Accounting period
- Financial year
- Journal status
- Source module
- Event type
- Branch
- Department
- Project
- Cost centre

22.2 Query Optimization

Avoid:

- Loading all journal lines into application memory.
- Repeated per-account queries.
- N+1 queries.
- Client-side aggregation of large ledgers.
- Full-table scans for period reports.

22.3 Reporting Caches

Materialized summaries may be introduced only when:

- The journal lines remain authoritative.
- Cache refresh rules are documented.
- Cache results are validated.
- Full cache reconstruction is possible.

22.4 Large Export Jobs

Large PDF and Excel exports should be processed safely without causing application timeouts or memory exhaustion.

23. PHASE 18 — PRODUCTION MIGRATION PLAN

23.1 Pre-Migration

- Complete production backup.
- Validate restoration.
- Deploy code to staging.
- Restore a recent production copy into staging.
- Run schema migrations.
- Run audit scripts.
- Run repair scripts.
- Rebuild reporting caches.
- Run integrity tests.
- Obtain finance-team sign-off.

23.2 Controlled Maintenance Window

During final migration:

- Restrict new accounting postings.
- Complete final backup.
- Record final transaction counts.
- Deploy code.
- Execute schema migrations.

- Execute approved data repair.
- Rebuild financial summaries.
- Run integrity checks.
- Re-enable postings only after validation.

23.3 Mandatory Production Validation

Confirm:

Total Debits = Total Credits

For:

- Every journal
- Every business
- Every accounting period
- Every financial year
- The complete platform

Also validate:

- Balance Sheets balance.
- Cash agrees with bank and cash ledgers.
- Receivables agree with customer balances.
- Payables agree with supplier balances.
- Equity agrees with equity reports.
- Historical reports are available.
- No duplicate journals were introduced.
- No production records were lost.

23.4 Rollback Conditions

Rollback if:

- Migrations fail.
- Journal totals change unexpectedly.
- Trial Balance becomes unbalanced.
- Production posting fails.
- Source links are lost.
- Critical reports fail.
- Cross-business data exposure is discovered.
- Migration performance threatens system stability.

24. POST-DEPLOYMENT MONITORING

Create automated integrity monitoring.

24.1 Daily Checks

- Unbalanced journals.
- Duplicate source postings.
- Missing journals.
- Journals without sources.
- Failed posting attempts.
- Closed-period violations.
- Inactive-account postings.
- Cross-business references.
- Report cache differences.

24.2 Monthly Checks

- Trial Balance.
- Balance Sheet equation.
- Receivable-control reconciliation.
- Payable-control reconciliation.
- Inventory reconciliation.
- Payroll-liability reconciliation.
- Bank reconciliation status.
- Equity reconciliation.
- Suspense accounts.
- Period-close readiness.

24.3 Accounting Integrity Dashboard

Display:

- Trial Balance status
 - Unbalanced journals
 - Duplicate posting warnings
 - Missing-account mappings
 - Reconciliation completion
 - Open-period status
 - Suspense-account balance
 - Unposted transactions
 - Historical repair warnings
 - Financial statement status
-

25. REQUIRED DEVELOPMENT DELIVERABLES

The implementation team must deliver:

1. Current-state accounting audit.
2. Target architecture document.
3. Complete posting matrix.
4. Chart of Accounts mapping registry.

5. Database schema changes.
 6. Migration scripts.
 7. Rollback scripts.
 8. Historical data-repair scripts.
 9. Accounting Posting Engine.
 10. Journal and reversal services.
 11. General Ledger service.
 12. Trial Balance service.
 13. Financial statement services.
 14. Accounting-period framework.
 15. Bank reconciliation module.
 16. Equity module.
 17. Period and year-end closing.
 18. Projection module.
 19. Loan-readiness module.
 20. Audit and permissions implementation.
 21. Automated test suite.
 22. Data-reconciliation reports.
 23. Deployment guide.
 24. Support and troubleshooting guide.
 25. Accounting architecture developer documentation.
-

26. PHASE ACCEPTANCE GATES

Each phase must pass a formal gate before the next one begins.

Gate A — Architecture

- All posting paths documented.
- Target architecture approved.
- No unknown financial mutation paths remain.

Gate B — Posting Engine

- Balanced journals guaranteed.
- Idempotency proven.
- Transaction rollback proven.
- Source linkage proven.

Gate C — General Ledger

- Ledger derived from journal lines.
- Opening and closing balances correct.
- Multi-business isolation confirmed.

Gate D — Reports

- Trial Balance balances.

- Balance Sheet balances.
- Income Statement reconciles.
- Cash Flow reconciles.
- Drill-down works.

Gate E — Historical Repair

- Duplicates classified.
- Wrong-account postings corrected.
- Owner capital discrepancy resolved.
- Liability discrepancy resolved.
- Repair logs complete.

Gate F — Period Controls

- Automatic period assignment works.
- Closed-period restrictions work.
- Reopening is audited.
- Closing checklist works.

Gate G — Banking and Equity

- Reconciliation adjustments post correctly.
- Completed reconciliations are auditable.
- Equity movements reconcile to the Balance Sheet.

Gate H — Production Readiness

- Full test suite passes.
- Production-like migration succeeds.
- Rollback succeeds.
- Finance team signs off.
- Auditor review is complete.

27. FINAL SYSTEM ACCEPTANCE CRITERIA

The implementation is complete only when all of the following are true:

1. Every posted journal balances.
2. Trial Balance total debits equal total credits.
3. The General Ledger agrees with the Trial Balance.
4. The Balance Sheet balances.
5. Net profit agrees with the movement in current-year earnings.
6. Cash Flow closing cash agrees with cash and bank ledgers.
7. Receivables agree with customer balances.
8. Payables agree with supplier balances.
9. Inventory agrees with the inventory control account.
10. Fixed assets agree with the asset register.

11. Loan balances agree with repayment schedules.
 12. Payroll liabilities agree with payroll records.
 13. Equity reports agree with equity accounts.
 14. Owner capital is not duplicated.
 15. Parent and child accounts are not double-counted.
 16. Duplicate transaction posting is technically prevented.
 17. Every report figure can be traced to journal lines.
 18. Every journal can be traced to a source transaction.
 19. Posted transactions cannot be edited silently.
 20. Closed periods protect historical records.
 21. Reversals and adjustments are fully auditable.
 22. Historical data is preserved.
 23. Migration scripts are reversible.
 24. Reports use the General Ledger as their only accounting source.
 25. Forecast data remains separate from actual accounting data.
 26. Business and tenant data are fully isolated.
 27. The system performs reliably with production-scale data.
 28. Finance personnel approve the corrected balances.
 29. Auditors can trace the complete transaction history.
 30. Production deployment completes without loss of data.
-

28. FINAL IMPLEMENTATION PRIORITY

The team must prioritize correctness over visual enhancement.

The implementation order must remain:

```
Accounting Integrity
  ↓
Historical Data Repair
  ↓
General Ledger
  ↓
Trial Balance
  ↓
Financial Statements
  ↓
Accounting Periods
  ↓
Operational Integrations
  ↓
Bank Reconciliation
  ↓
Equity Management
  ↓
Forecasting
```


↓
Executive Dashboards

No dashboard, report design, AI forecast, or visual feature should be considered reliable until the underlying accounting engine has passed all integrity checks.

The final result must be an accounting backend in which every value is posted once, mapped correctly, balanced, period-controlled, immutable after posting, and traceable from the financial statements to the original business transaction.